

Eleventh Section.

The inflection points of a curve of the third order.

§ 85. Ternary forms and algebraic curves.

In order, in the further course of the exposition, to present some of the many interesting applications of algebra to geometrical problems, without referring the reader to other aids, the necessary geometrical foundations for these applications are here derived briefly. We restrict ourselves to plane geometry and even there leave aside everything which is not directly required for the applications.¹

We use homogeneous triangular coordinates. The position of a point x is determined by its distances from the three sides of the coordinate triangle, measured in three arbitrary units or multiplied by three arbitrary constant factors. These quantities are the coordinates of the point x , and are denoted by x_1, x_2, x_3 or by y_1, y_2, y_3 . Between them there is at first a linear non-homogeneous relation. By means of this relation the unit can be represented linearly and homogeneously by x_1, x_2, x_3 , and every non-homogeneous function can be changed into a homogeneous one.

The coordinates x_1, x_2, x_3 are now regarded as independent variables, provided only homogeneous forms and equations are used. Not x_1, x_2, x_3 themselves, but only the ratios $x_1 : x_2 : x_3$, determine the point. The value combination $x_1 = x_2 = x_3 = 0$ is excluded; hx_1, hx_2, hx_3 determine the same point for every $h \neq 0$.

Every linear substitution

$$\begin{aligned} x_1 &= \alpha_1^{(1)} y_1 + \alpha_1^{(2)} y_2 + \alpha_1^{(3)} y_3, \\ x_2 &= \alpha_2^{(1)} y_1 + \alpha_2^{(2)} y_2 + \alpha_2^{(3)} y_3, \\ x_3 &= \alpha_3^{(1)} y_1 + \alpha_3^{(2)} y_2 + \alpha_3^{(3)} y_3, \end{aligned} \quad (1)$$

whose determinant

$$r = \sum \pm \alpha_1^{(1)} \alpha_2^{(2)} \alpha_3^{(3)} \quad (2)$$

is not zero, admits a twofold geometrical interpretation: either as a mapping of two points in the same coordinate system, or as a coordinate transformation of one and the same point. In the next considerations the latter interpretation is retained.

An arbitrary ternary form of degree n , $f(x_1, x_2, x_3)$, when set equal to zero, represents a curve of order n ; a straight line has n intersections with it. By (1) the form f passes into a form of the same degree in y_1, y_2, y_3 ,

$$f(x_1, x_2, x_3) = \varphi(y_1, y_2, y_3), \quad (3)$$

and $\varphi = 0$ represents the same curve in the new coordinates. Differentiating (3) with respect to y_1, y_2, y_3 gives

$$\begin{aligned} \varphi'(y_1) &= \alpha_1^{(1)} f'(x_1) + \alpha_2^{(1)} f'(x_2) + \alpha_3^{(1)} f'(x_3), \\ \varphi'(y_2) &= \alpha_1^{(2)} f'(x_1) + \alpha_2^{(2)} f'(x_2) + \alpha_3^{(2)} f'(x_3), \\ \varphi'(y_3) &= \alpha_1^{(3)} f'(x_1) + \alpha_2^{(3)} f'(x_2) + \alpha_3^{(3)} f'(x_3). \end{aligned} \quad (4)$$

§ 86. Singular points. Inflection points. Bitangents.

If for a point x the three equations

$$f'(x_1) = 0, \quad f'(x_2) = 0, \quad f'(x_3) = 0 \quad (1)$$

are simultaneously satisfied, then by Euler's theorem

$$nf = x_1 f'(x_1) + x_2 f'(x_2) + x_3 f'(x_3) \quad (2)$$

¹More extensive material is found in textbooks of analytic geometry, among which George Salmon's *Higher plane curves*, German by Fiedler, should be noted.

the point lies on the curve f . The equations § 85, (4) also show that this property is preserved under linear transformations. Such points do not occur on every curve of order n . Eliminating x_1, x_2, x_3 from (1) gives a relation among the coefficients of f ; this integral rational homogeneous function of the coefficients is called the discriminant of f , and is determined only up to a numerical factor. Points of the curve whose coordinates satisfy (1) are called singular points.

If a curve is to have infinitely many singular points, then the partial derivatives must have a common non-constant divisor; by the same considerations this divisor must also have a common divisor with f . A curve with infinitely many singular points is therefore reducible, and the points lie on a multiple component of the curve.

For the geometrical interpretation, order f according to powers of x_3 :

$$f(x_1, x_2, x_3) = x_3^n f_0 + x_3^{n-1} f_1 + x_3^{n-2} f_2 + \cdots, \quad (3)$$

where $f_0, f_1, f_2,$

$\dots$ are binary forms in x_1, x_2 and the degree is indicated by the subscript. If a singular point is at the vertex I_3 , then f_0 and f_1 must vanish, and the lines through I_3 tangent to the curve are obtained from $f_2 = 0$. If f_2 is not identically zero, I_3 is a double point; if also $f_2 = 0$, one obtains a singular point of higher multiplicity.

A point of a curve is called an inflection point if the tangent has three-point contact with the curve there. If I_3 is an inflection point and $x_1 = 0$ is the inflection tangent, then f has the form

$$f = x_1 \varphi + x_3^2 \psi, \quad (4)$$

where φ has degree $n - 1$ and ψ degree $n - 2$. Stronger divisibility expresses four-point contact. A straight line which touches the curve at two distinct points is called a bitangent. If it is chosen as $x_3 = 0$ and the points of contact are chosen as the two vertices on $x_3 = 0$, then $f|_{x_3=0}$ must have both x_1^2 and x_2^2 as factors.

§ 87. Covariants of ternary forms.

Let

$$f(x_1, x_2, x_3) = \sum a_{i_1 i_2 i_3} x_1^{i_1} x_2^{i_2} x_3^{i_3}$$

be a ternary form. A form $C(x, a)$, homogeneous in the variables x and in the coefficients a , is called a covariant of f if under the transformation (1) of the variables it transforms by a single factor:

$$C(y, b) = r^\lambda C(x, a). \quad (1)$$

If the form is independent of the variables, it is an invariant. The exponent λ is the weight.

For every ternary form of degree greater than 1 there is the Hessian determinant. With the normalization used here we write

$$H(x) = \det(f_{ij})_{1 \leq i, j \leq 3}, \quad f_{ij} = f''(x_i, x_j), \quad (2)$$

or, up to Weber's chosen numerical factor, $\Delta(x)$. Besides this covariant Weber considers two further general covariants. First, from the second derivatives of f and the first derivatives of the Hessian covariant, one forms

$$J(x) = \frac{1}{36} \begin{vmatrix} f_{11} & f_{12} & f_{13} & \Delta_1 \\ f_{21} & f_{22} & f_{23} & \Delta_2 \\ f_{31} & f_{32} & f_{33} & \Delta_3 \\ \Delta_1 & \Delta_2 & \Delta_3 & 0 \end{vmatrix}, \quad (3)$$

where $\Delta_i = \Delta'(x_i)$. Then the third covariant is the functional determinant of f, Δ, J :

$$K(x) = \frac{1}{9} \begin{vmatrix} f_1 & \Delta_1 & J_1 \\ f_2 & \Delta_2 & J_2 \\ f_3 & \Delta_3 & J_3 \end{vmatrix}. \quad (4)$$

Their transformation laws are

$$\Delta(y) = r^2 \Delta(x), \quad J(y) = r^6 J(x), \quad K(y) = r^9 K(x), \quad (5)$$

which gives the corresponding weights.

§ 88. The Hessian curve.

The curve represented by $H = 0$ is called the Hessian curve of the curve $f = 0$. To see its relation with inflection points, write f at the point I_3 in the form

$$f = x_3^n f_0 + x_3^{n-1} f_1 + x_3^{n-2} f_2 + \cdots. \quad (1)$$

If one inserts the form corresponding to an inflection point I_3 with inflection tangent $x_1 = 0$, then the Hessian covariant, ordered by descending powers of x_3 , becomes

$$H = x_3^{3n-6} H_0 + x_3^{3n-7} H_1 + \cdots, \quad (2)$$

with

$$H_0 = -2(n-1)^2 h^2 c. \quad (3)$$

Thus the Hessian curve passes through I_3 precisely when $c = 0$, that is, precisely when I_3 is an inflection point of the curve f . From the expression for H_1 one further reads off when the inflection tangent is also tangent to the Hessian curve. For a curve of the third order this four-point contact can occur only if the cubic form becomes reducible.

Thus, for a nonsingular cubic curve, the determination of the inflection points is reduced to the intersections of $f = 0$ and $H = 0$. Since both curves are cubic, there are nine inflection points, apart from exceptional cases.

§ 89. Inflection points of a curve of the third order.

Let $f = 0$ be the equation of a curve of the third order without singular point. Place two vertices of the coordinate triangle, I_1, I_2 , in two of the nine inflection points, and choose the corresponding inflection tangents as the sides $x_1 = 0$ and $x_2 = 0$. Then, when $x_1 = 0$ or $x_2 = 0$ is put, f must reduce to one term hx_3^3 . Hence f has the form

$$f(x_1, x_2, x_3) = x_1 x_2 (a_1 x_1 + a_2 x_2 + a_3 x_3) + h x_3^3. \quad (1)$$

The straight line

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = 0 \quad (2)$$

is likewise an inflection tangent, and its intersection with $x_3 = 0$ is the third inflection point on this line.

Let ε be an imaginary cube root of unity. Introduce two linear functions z_1, z_2 by

$$\begin{aligned} a_1 x_1 &= \varepsilon z_1 + \varepsilon^2 z_2 - \frac{1}{3} a_3 x_3, \\ a_2 x_2 &= \varepsilon^2 z_1 + \varepsilon z_2 - \frac{1}{3} a_3 x_3, \end{aligned} \quad (3)$$

from which also

$$-(a_1 x_1 + a_2 x_2 + a_3 x_3) = z_1 + z_2 - \frac{1}{3} a_3 x_3 \quad (4)$$

follows. On substituting in (1), and after multiplying by suitable constant factors, one obtains the symmetric representation

$$f = \varphi(y_1, y_2, y_3) = h(y_1^3 + y_2^3 + y_3^3) + 6m y_1 y_2 y_3. \quad (5)$$

Consequently a nonsingular ternary cubic form can be transformed in four different ways into the canonical form (5). Since on each of the lines $y_1 = 0, y_2 = 0, y_3 = 0$ there are three inflection points, the problem of the inflection points is essentially the same as the transformation of the cubic form to canonical form.²

§ 90. Transformation of the cubic form to canonical form.

For the canonical form write

$$\varphi(y) = h \sum y_i^3 + 6m y_1 y_2 y_3. \quad (1)$$

To determine the transformation one first computes the covariants of the canonical form. The Hessian covariant is

$$\Delta(y) = -hm^2 \sum y_i^3 + (h^3 + 2m^3) y_1 y_2 y_3. \quad (2)$$

The second covariant is written in the normalization

$$J(x) = \frac{1}{36} \begin{vmatrix} f_{11} & f_{12} & f_{13} & \Delta_1 \\ f_{21} & f_{22} & f_{23} & \Delta_2 \\ f_{31} & f_{32} & f_{33} & \Delta_3 \\ \Delta_1 & \Delta_2 & \Delta_3 & 0 \end{vmatrix}, \quad (3)$$

and for the canonical form

$$\begin{aligned} J(y) = & -h^2(h^3 + 8m^3)^2 \sum y_i^3 y_j^3 + 3m^2(5h^6 + 26h^3 m^3 - 4m^6) y_1^2 y_2^2 y_3^2 \\ & + hm(2h^6 + 5h^3 m^3 + 20m^6) y_1 y_2 y_3 \sum y_i^3 + 9h^2 m^6 \left(\sum y_i^3 \right)^2. \end{aligned} \quad (4)$$

From the invariant relations

$$\varphi(y) = f, \quad \Delta(y) = r^2 \Delta, \quad J(y) = r^6 J \quad (5)$$

one obtains first

$$(h^3 + 8m^3) y_1 y_2 y_3 = m^2 f + r^2 \Delta, \quad (6)$$

$$h(h^3 + 8m^3) \sum y_i^3 = (h^3 + 2m^3) f - 6mr^2 \Delta. \quad (7)$$

Then (4) gives

$$\begin{aligned} h^2(h^3 + 8m^3)^2 \sum y_i^3 y_j^3 = & (2h^3 + m^3) m^3 f^2 + m(2h^3 - 5m^3) r^2 f \Delta \\ & + 3m^2 r^4 \Delta^2 - r^6 J. \end{aligned} \quad (8)$$

The third covariant is

$$K(x) = \frac{1}{9} \begin{vmatrix} f_1 & \Delta_1 & J_1 \\ f_2 & \Delta_2 & J_2 \\ f_3 & \Delta_3 & J_3 \end{vmatrix}, \quad (9)$$

and for the canonical form

$$K(y) = -h^3(h^3 + 8m^3)^3 \begin{vmatrix} 1 & y_1^3 & y_1^6 \\ 1 & y_2^3 & y_2^6 \\ 1 & y_3^3 & y_3^6 \end{vmatrix}. \quad (10)$$

Since $K(y) = r^9 K$, it follows that

$$r^9 K = h^3(h^3 + 8m^3)^3 (y_1^3 - y_2^3)(y_1^3 - y_3^3)(y_2^3 - y_3^3). \quad (11)$$

²Weber refers here to Newton and Maclaurin on inflection points of cubic curves, Hesse on elimination, and Aronhold on homogeneous functions of the third degree.

If there is no singular point, $h^3 + 8m^3$ cannot vanish. Put $h = 1$. Then (6), (7), (8) give the cubic equation

$$u^3 - Qu^2 + Pu - R^3 = 0, \quad (12)$$

whose roots are y_1^3, y_2^3, y_3^3 , with

$$P = \frac{(2 + m^3)m^3f^2 + (2 - 5m^3)mr^2f\Delta + 3m^2r^4\Delta^2 - r^6J}{(1 + 8m^3)^2}, \quad (13)$$

$$Q = \frac{(1 + 2m^3)f - 6mr^2\Delta}{1 + 8m^3}, \quad (14)$$

$$R = \frac{m^2f + r^2\Delta}{1 + 8m^3}. \quad (15)$$

The discriminant of this cubic equation is

$$D_1 = \frac{r^{18}K^2}{(1 + 8m^3)^6}. \quad (16)$$

§ 91. Invariants of the curve of the third order. Reality.

For the computation of the invariants Weber considers the determinant of the Hessian covariant of the form

$$\Delta_{\lambda,\mu} = \lambda f + \mu \Delta. \quad (1)$$

It can be represented in the form

$$\Delta_{\lambda,\mu}(x) = Lf + M\Delta, \quad (2)$$

where L, M are binary forms of the third degree in λ, μ . For the canonical form, replacing in § 90, (2) the quantities

$$h, \quad m$$

by

$$h(\lambda - 6m^2\mu), \quad m\lambda + (h^3 + 2m^3)\mu$$

gives

$$\begin{aligned} \Delta_{\lambda,\mu} = & -h(\lambda - 6m^2\mu)[m\lambda + (h^3 + 2m^3)\mu]^2 \sum y_i^3 \\ & + \{h^3(\lambda - 6m^2\mu)^3 + 2[m\lambda + (h^3 + 2m^3)\mu]^3\} y_1 y_2 y_3. \end{aligned} \quad (3)$$

Consequently

$$\begin{aligned} L = & -2\lambda^2\mu(h^3 - m^3)m - \lambda\mu^2(h^6 - 20h^3m^3 - 8m^6) + 8\mu^3m^2(h^3 - m^3)^2, \\ M = & \lambda^3 + 12\lambda^2\mu m(h^3 - m^3) + 2\mu^3(h^6 - 20h^3m^3 - 8m^6). \end{aligned} \quad (4)$$

Thus one obtains two invariants of degrees four and six, in canonical form

$$m(h^3 - m^3), \quad h^6 - 20h^3m^3 - 8m^6. \quad (5)$$

Denote the corresponding invariants in the general form by S, T , and put $h = 1$. Then

$$m(1 - m^3) = r^4 S, \quad 1 - 20m^3 - 8m^6 = r^6 T. \quad (6)$$

For L, M one obtains

$$L = -2S\lambda^2\mu - T\lambda\mu^2 + 8S^2\mu^3, \quad M = \lambda^3 + 12S\lambda^2\mu + 2T\mu^3. \quad (7)$$

Eliminating m^2 and m^5 gives the equation

$$27m^8 + 18Sm^4r^4 + Tm^2r^6 - S^2r^8 = 0. \quad (8)$$

With

$$z = \frac{3m^2}{r^2} \quad (9)$$

this becomes

$$z^4 + 6Sz^2 + Tz - 3S^2 = 0. \quad (10)$$

The discriminant is

$$D_2 = -27(T^2 + 64S^3)^2 = -27D^2, \quad (11)$$

where $D = T^2 + 64S^3$. For the canonical form

$$D = h^3(h^3 + 8m^3)^3, \quad (12)$$

and hence D cannot vanish as long as the curve has no singular point. It follows that for a real curve the biquadratic equation has negative discriminant, and therefore two real and two conjugate imaginary roots. For the positive real root z , the corresponding values m and r are real, and consequently also y_1, y_2, y_3 are real. Thus:

Every real ternary cubic form with non-vanishing discriminant can be brought, by a real transformation, into the real canonical form.

In canonical coordinates the nine inflection points are obtained from

$$y_1^3 + y_2^3 + y_3^3 + 6my_1y_2y_3 = 0 \quad (13)$$

by putting successively y_1, y_2, y_3 equal to zero. If ε is an imaginary cube root of unity, their coordinates are

$$\begin{array}{lll} (0, 1, -1), & (0, 1, -\varepsilon), & (0, 1, -\varepsilon^2), \\ (-1, 0, 1), & (-\varepsilon, 0, 1), & (-\varepsilon^2, 0, 1), \\ (1, -1, 0), & (1, -\varepsilon, 0), & (1, -\varepsilon^2, 0). \end{array} \quad (14)$$

The three points in each row lie on the lines $y_1 = 0, y_2 = 0, y_3 = 0$. The remaining triples of collinear points are arranged as

$$\begin{array}{lll|lll|lll} (0, 1, -1) & (-1, 0, 1) & (1, -1, 0) & (0, 1, -1) & (-\varepsilon, 0, 1) & (1, -\varepsilon^2, 0) & (0, 1, -1) & (-\varepsilon^2, 0, 1) & (1, -\varepsilon, 0) \\ (0, 1, -\varepsilon) & (-\varepsilon, 0, 1) & (1, -\varepsilon, 0) & (0, 1, -\varepsilon) & (-\varepsilon^2, 0, 1) & (1, -1, 0) & (0, 1, -\varepsilon) & (-1, 0, 1) & (1, -\varepsilon^2, 0) \\ (0, 1, -\varepsilon^2) & (-\varepsilon^2, 0, 1) & (1, -\varepsilon^2, 0) & (0, 1, -\varepsilon^2) & (-1, 0, 1) & (1, -\varepsilon, 0) & (0, 1, -\varepsilon^2) & (-\varepsilon, 0, 1) & (1, -1, 0) \end{array} \quad (15)$$

For the reality relations, besides (14), the conjugate systems are

$$\begin{array}{lll} (0, 1, -1), & (-\varepsilon, 0, 1), & (1, -\varepsilon^2, 0) \\ (-1, 0, 1), & (1, -\varepsilon, 0), & (0, 1, -\varepsilon^2) \\ (1, -1, 0), & (0, 1, -\varepsilon), & (-\varepsilon^2, 0, 1), \end{array} \quad (16)$$

and

$$\begin{array}{lll} (0, 1, -1), & (-\varepsilon^2, 0, 1), & (1, -\varepsilon, 0) \\ (-1, 0, 1), & (1, -\varepsilon^2, 0), & (0, 1, -\varepsilon) \\ (1, -1, 0), & (0, 1, -\varepsilon^2), & (-\varepsilon, 0, 1). \end{array} \quad (17)$$

A still clearer notation is obtained by letting ξ and η each run through a complete residue system modulo 3. The nine inflection points are denoted by the pairs (ξ, η) ; (ξ, η) and (η, ξ) are conjugate imaginary points, and $(0, 0), (1, 1), (2, 2)$ are real. The four systems of lines are

$$\begin{array}{lll|lll} (0, 0) & (1, 2) & (2, 1) & (0, 0) & (1, 1) & (2, 2) \\ (1, 1) & (2, 0) & (0, 2) & (1, 2) & (2, 0) & (0, 1) \\ (2, 2) & (0, 1) & (1, 0) & (2, 1) & (0, 2) & (1, 0) \\ (0, 0) & (2, 0) & (1, 0) & (0, 0) & (0, 2) & (0, 1) \\ (1, 1) & (0, 1) & (2, 1) & (1, 1) & (1, 0) & (1, 2) \\ (2, 2) & (1, 2) & (0, 2) & (2, 2) & (2, 1) & (2, 0). \end{array} \quad (18)$$

Therefore three inflection points (ξ_1, η_1) , (ξ_2, η_2) , (ξ_3, η_3) lie on a straight line if and only if

$$\xi_1 + \xi_2 + \xi_3 \equiv 0, \quad \eta_1 + \eta_2 + \eta_3 \equiv 0 \pmod{3}. \quad (19)$$

The notation of the four systems of straight lines can also be derived from the schema

$$\begin{array}{ccc} (0, 0) & (1, 2) & (2, 1) \\ (1, 1) & (2, 0) & (0, 2) \\ (2, 2) & (0, 1) & (1, 0), \end{array} \quad (20)$$

by taking the rows, the columns, and the triples corresponding to the positive and negative terms of the determinant.